COMP 5212

Machine Learning

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Lecture 1: Math Basics

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Adapted from the slides of Prof. Junxian He

Outline

- Linear Algebra Review
- Probability Review

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- Linear Algebra Review
- Probability Review

Basic Notation

- By $x \in \mathbb{R}^n$, we denote a vector with n entries (we use column vectors by default).

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

- By $A \in \mathbb{R}^{m \times n}$ we denote a matrix with m rows and n columns, where the entries of A are real numbers.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} a^1 & a^2 & \cdots & a^n \end{bmatrix} = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix}.$$

Identity Matrix

The *identity matrix*, denoted $I \in \mathbb{R}^{n \times n}$, is a square matrix with ones on the diagonal and zeros everywhere else. That is,

$$I_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

It has the property that for all $A \in \mathbb{R}^{m \times n}$,

$$AI = A = IA.$$

Diagonal Matrix

A *diagonal matrix* is a matrix where all non-diagonal elements are 0. This is typically denoted

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n),$$

with

$$D_{ij} = \begin{cases} d_i & i = j, \\ 0 & i \neq j. \end{cases}$$

Clearly, $I = \operatorname{diag}(1, 1, \ldots, 1)$.

Vector-Vector Product

- *inner product* or *dot product*

$$x^T y \in \mathbb{R} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^{n} x_i y_i .$$

- *outer product*

$$xy^T \in \mathbb{R}^{m \times n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{bmatrix} .$$

Matrix-Vector Product

- If we write A by rows, then we can express Ax as,

$$y = Ax = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix} x = \begin{bmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{bmatrix}.$$

Matrix-Vector Product

- If we write A by columns, then we have:

$$y = Ax = \begin{bmatrix} a^1 & a^2 & \cdots & a^n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = a^1 x_1 + a^2 x_2 + \cdots + a^n x_n. \quad (1)$$

y is a *linear combination* of the columns of A.

Matrix-Vector Product

It is also possible to multiply on the left by a row vector.

- If we write A by columns, then we can express $x^T A$ as,

$$y^T = x^T A = x^T \begin{bmatrix} a^1 & a^2 & \cdots & a^n \end{bmatrix} = \begin{bmatrix} x^T a^1 & x^T a^2 & \cdots & x^T a^n \end{bmatrix}.$$

Matrix-Matrix Multiplication

1. As a set of vector-vector products (dot product)

$$C = AB = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix} \begin{bmatrix} b^1 & b^2 & \cdots & b^p \end{bmatrix} = \begin{bmatrix} a_1^T b^1 & a_1^T b^2 & \cdots & a_1^T b^p \\ a_2^T b^1 & a_2^T b^2 & \cdots & a_2^T b^p \\ \vdots & \vdots & \ddots & \vdots \\ a_m^T b^1 & a_m^T b^2 & \cdots & a_m^T b^p \end{bmatrix}.$$

Matrix-Matrix Multiplication

2. As a sum of outer products

$$C = AB = \begin{bmatrix} a^1 & a^2 & \cdots & a^p \end{bmatrix} \begin{bmatrix} b_1^T \\ b_2^T \\ \vdots \\ b_p^T \end{bmatrix} = \sum_{i=1}^{p} a^i b_i^T .$$

Matrix-Matrix Multiplication

3. As a set of matrix-vector products.

$$C = AB = A \begin{bmatrix} b^1 & b^2 & \cdots & b^n \end{bmatrix} = \begin{bmatrix} Ab^1 & Ab^2 & \cdots & Ab^n \end{bmatrix}. \tag{2}$$

Here the ith column of C is given by the matrix-vector product with the vector on the right, $c_i = Ab_i$. These matrix-vector products can in turn be interpreted using both viewpoints given in the previous subsection.

Matrix-Matrix Multiplication

4. As a set of vector-matrix products.

$$C = AB = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix} B = \begin{bmatrix} a_1^T B \\ a_2^T B \\ \vdots \\ a_m^T B \end{bmatrix}.$$

Matrix-Matrix Multiplication

- Associative: $(AB)C = A(BC)$.
- Distributive: $A(B + C) = AB + AC$.
- In general, *not* commutative; that is, it can be the case that $AB \neq BA$. (For example, if $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times q}$, the matrix product BA does not even exist if m and q are not equal!)

The Transpose

The *transpose* of a matrix results from "flipping" the rows and columns. Given a matrix $A \in \mathbb{R}^{m \times n}$, its transpose, written $A^T \in \mathbb{R}^{n \times m}$, is the $n \times m$ matrix whose entries are given by

$$(A^T)_{ij} = A_{ji}.$$

The following properties of transposes are easily verified:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $(A + B)^T = A^T + B^T$

Trace

The *trace* of a square matrix $A \in \mathbb{R}^{n \times n}$, denoted $\operatorname{tr} A$, is the sum of diagonal elements in the matrix:

$$\operatorname{tr} A = \sum_{i=1}^{n} A_{ii}.$$

Norms

A *norm* of a vector $\|x\|$ is informally a measure of the "length" of the vector.

The commonly-used Euclidean or ℓ_2 norm,

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

The ℓ_1 norm,

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|.$$

Norms

A *norm* of a vector $\|x\|$ is informally a measure of the "length" of the vector.

The ℓ_∞ norm,

$$\|x\|_\infty = \max_i |x_i|.$$

Norms

In fact, all three norms presented so far are examples of the family of ℓ_p norms, which are parameterized by a real number $p \geq 1$, and defined as

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$

Linear Independence

A set of vectors $\{x_1, x_2, \ldots, x_n\} \subset \mathbb{R}^m$ is said to be (linearly) *dependent* if one vector belonging to the set can be represented as a linear combination of the remaining vectors; that is, if

$$x_n = \sum_{i=1}^{n-1} \alpha_i x_i$$

for some scalar values $\alpha_1, \ldots, \alpha_{n-1} \in \mathbb{R}$; otherwise, the vectors are (linearly) *independent*.

Linear Independence

Example:

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}, \qquad x_3 = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$$

are linearly dependent because $x_3 = -2x_1 + x_2$.

Rank of a Matrix

- The *column rank* of a matrix $A \in \mathbb{R}^{m \times n}$ is the largest number of columns of A that constitute a linearly independent set.

- The *row rank* is the largest number of rows of A that constitute a linearly independent set.

- For any matrix $A \in \mathbb{R}^{m \times n}$, it turns out that the column rank of A is equal to the row rank of A (prove it yourself!), and so both quantities are referred to collectively as the *rank* of A, denoted as $\text{rank}(A)$.

Properties of Rank

- For $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(A) \leq \min(m, n)$. If $\operatorname{rank}(A) = \min(m, n)$, then A is said to be *full rank*.
- For $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(A) = \operatorname{rank}(A^T)$.
- For $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{p \times n}$, $\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$.
- For $A, B \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(A + B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.

The Inverse of a Square Matrix

- The *inverse* of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted A^{-1}, and is the unique matrix such that

$$A^{-1}A = I = AA^{-1}.$$

- We say that A is *invertible* or *non-singular* if A^{-1} exists and *non-invertible* or *singular* otherwise.
- In order for a square matrix A to have an inverse A^{-1}, then A must be full rank.
- Properties (Assuming $A, B \in \mathbb{R}^{n \times n}$ are non-singular):
    - $(A^{-1})^{-1} = A$
    - $(AB)^{-1} = B^{-1}A^{-1}$
    - $(A^{-1})^T = (A^T)^{-1}$. For this reason this matrix is often denoted A^{-T}.

Orthogonal Matrices

- Two vectors $x, y \in \mathbb{R}^n$ are *orthogonal* if $x^T y = 0$.
- A vector $x \in \mathbb{R}^n$ is *normalized* if $\|x\|_2 = 1$.
- A square matrix $U \in \mathbb{R}^{n \times n}$ is *orthogonal* if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being *orthonormal*).

Properties:

- The inverse of an orthogonal matrix is its transpose.

$$U^T U = I = U U^T.$$

- Operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,

$$\|Ux\|_2 = \|x\|_2$$

for any $x \in \mathbb{R}^n$, $U \in \mathbb{R}^{n \times n}$ orthogonal.

Span and Projection

- The *span* of a set of vectors $\{x_1, x_2, \ldots, x_n\}$ is the set of all vectors that can be expressed as a linear combination of $\{x_1, \ldots, x_n\}$. That is,

$$\text{span}(\{x_1, \ldots, x_n\}) = \left\{ v : v = \sum_{i=1}^{n} \alpha_i x_i, \;\; \alpha_i \in \mathbb{R} \right\}.$$

- The *projection* of a vector $y \in \mathbb{R}^m$ onto the span of $\{x_1, \ldots, x_n\}$ is the vector $v \in \text{span}(\{x_1, \ldots, x_n\})$, such that v is as close as possible to y, as measured by the Euclidean norm $\|v - y\|_2$.

$$\text{Proj}(y; \{x_1, \ldots, x_n\}) = \underset{v \in \text{span}(\{x_1, \ldots, x_n\})}{\arg\min} \|y - v\|_2.$$

Range

- The *range* or the column space of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{R}(A)$, is the span of the columns of A. In other words,

$$\mathcal{R}(A) = \{v \in \mathbb{R}^m : v = Ax,\ x \in \mathbb{R}^n\}.$$

- Assuming A is full rank and $n < m$, the projection of a vector $y \in \mathbb{R}^m$ onto the range of A is given by,

$$\mathrm{Proj}(y; A) = \operatorname*{arg\,min}_{v \in \mathcal{R}(A)} \|v - y\|_2.$$

Null Space

The *nullspace* of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{N}(A)$ is the set of all vectors that equal 0 when multiplied by A, i.e.,

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

Determinant

Let $A \in \mathbb{R}^{n \times n}$, $A_{\setminus i, \setminus j} \in \mathbb{R}^{(n-1)\times(n-1)}$ be the matrix that results from deleting the ith row and jth column from A.
The general (recursive) formula for the determinant is

$$|A| = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \qquad \text{(for any } j \in \{1, \dots, n\}\text{)}$$

$$= \sum_{j=1}^{n} (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \qquad \text{(for any } i \in \{1, \dots, n\}\text{)}.$$

Determinant: Example

However, the equations for determinants of matrices up to size 3×3 are fairly common, and it is good to know them:

$$\left|\begin{bmatrix} a_{11} \end{bmatrix}\right| = a_{11}$$

$$\left|\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right| = a_{11}a_{22} - a_{12}a_{21}$$

$$\begin{aligned} \left|\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\right| &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ &\quad - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}. \end{aligned}$$

The Determinant

The *determinant* of a square matrix $A \in \mathbb{R}^{n \times n}$, is a function $\det : \mathbb{R}^{n \times n} \to \mathbb{R}$, and is denoted $|A|$ or $\det A$.

Given a matrix

$$A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix},$$

consider the set of points $S \subset \mathbb{R}^n$ as follows:

$$S = \left\{ v \in \mathbb{R}^n : v = \sum_{i=1}^{n} \alpha_i a_i \text{ where } 0 \leq \alpha_i \leq 1,\ i = 1, \ldots, n \right\}.$$

The absolute value of the determinant of A is a measure of the "volume" of the set S.

The Determinant

For example, consider the 2×2 matrix,

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}. \tag{3}$$

Here, the rows of the matrix are

$$a_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad a_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

The Determinant: Properties

Algebraically, the determinant satisfies the following three properties:

1. The determinant of the identity is 1, $|I| = 1$. (Geometrically, the volume of a unit hypercube is 1).

2. Given a matrix $A \in \mathbb{R}^{n \times n}$, if we multiply a single row in A by a scalar $t \in \mathbb{R}$, then the determinant of the new matrix is $t|A|$. (Geometrically, multiplying one of the sides of the set S by a factor t causes the volume to increase by a factor t.)

3. If we exchange any two rows a_i^T and a_j^T of A, then the determinant of the new matrix is $-|A|$, for example

The Determinant: Properties

- For $A \in \mathbb{R}^{n\times n}$, $|A| = |A^T|$.

- For $A, B \in \mathbb{R}^{n\times n}$, $|AB| = |A||B|$.

- For $A \in \mathbb{R}^{n\times n}$, $|A| = 0$ if and only if A is singular (i.e., non-invertible). (If A is singular then it does not have full rank, and hence its columns are linearly dependent. In this case, the set S corresponds to a "flat sheet" within the n-dimensional space and hence has zero volume.)

- For $A \in \mathbb{R}^{n\times n}$ and A non-singular, $|A^{-1}| = 1/|A|$.

Eigenvalues and Eigenvectors

Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say that $\lambda \in \mathbb{C}$ is an *eigenvalue* of A and $x \in \mathbb{C}^n$ is the corresponding *eigenvector* if

$$Ax = \lambda x, \qquad x \neq 0.$$

Intuitively, this definition means that multiplying A by the vector x

results in a new vector that points in the same direction as x, but scaled by a factor λ.

Gradient over Matrix

Suppose that $f : \mathbb{R}^{m\times n} \to \mathbb{R}$ is a function that takes as input a matrix A of size $m \times n$ and returns a real value. Then the gradient of f (with respect to $A \in \mathbb{R}^{m\times n}$) is the matrix of partial derivatives, defined as:

$$\nabla_A f(A) \in \mathbb{R}^{m\times n} = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \cdots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \cdots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \cdots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix}.$$

i.e., an $m \times n$ matrix with

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}.$$

Gradient over Vector

Note that the size of $\nabla_A f(A)$ is always the same as the size of A. So if, in particular, A is just a vector $x \in \mathbb{R}^n$,

$$\nabla_x f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}.$$

It follows directly from the equivalent properties of partial derivatives that:

- $\nabla_x (f(x) + g(x)) = \nabla_x f(x) + \nabla_x g(x)$.
- For $t \in \mathbb{R}$, $\nabla_x (t f(x)) = t \nabla_x f(x)$.

The Hessian

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is a function that takes a vector in $\mathbb{R}^n$ and returns a real number. Then the *Hessian* matrix with respect to x, written $\nabla_x^2 f(x)$ or simply as H is the $n \times n$ matrix of partial derivatives,

$$\nabla_x^2 f(x) \in \mathbb{R}^{n\times n} = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}.$$

Note that the Hessian is always symmetric, since

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}.$$

Gradients of Linear Functions

For $x \in \mathbb{R}^n$, let $f(x) = b^T x$ for some known vector $b \in \mathbb{R}^n$. Then

$$f(x) = \sum_{i=1}^{n} b_i x_i$$

so

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^{n} b_i x_i = b_k.$$

From this we can easily see that $\nabla_x b^T x = b$. This should be compared to the analogous situation in single variable calculus, where $\partial(ax)/\partial x = a$.

Common Gradient Formula

- $\nabla_x b^T x = b$
- $\nabla_x^2 b^T x = 0$
- $\nabla_x x^T A x = 2Ax$ (if A symmetric)
- $\nabla_x^2 x^T A x = 2A$ (if A symmetric)

Least Squares

- Given a full rank matrix $A \in \mathbb{R}^{m \times n}$, and a vector $b \in \mathbb{R}^m$ such that $b \notin \mathcal{R}(A)$, we want to find a vector x such that Ax is as close as possible to b, as measured by the square of the Euclidean norm $\|Ax - b\|_2^2$.

Outline

- Linear Algebra Review
- Probability Review

Basic Concepts

- Performing an **experiment** $\longrightarrow$ an **outcome**
- Sample space (S): the set of all possible outcomes of an experiment
- Event (E): a subset of S, that is, $E \subseteq S$

- **Non-negativity:** $P(E) \geq 0$ for every $E \subseteq S$.
- **Normalization:** $P(S) = 1$.
- **Countable additivity:** For every sequence of pairwise disjoint events $E_1, E_2, \ldots$,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

If $E \subseteq F$, then $P(E) \leq P(F)$.
The inclusion-exclusion principle for two events is

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

General inclusion-exclusion principle:

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{r=1}^{n} (-1)^{r+1} \sum_{1 \leq i_1 < \cdots < i_r \leq n} P\left(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_r}\right).$$

Conditional Probability and Bayes' Rule

For any events A, B such that $P(B) \neq 0$, we define:

$$P(A \mid B) := \frac{P(A \cap B)}{P(B)}.$$

Let's apply conditional probability to obtain Bayes' Rule!

$$\begin{aligned} P(B \mid A) &= \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)} \\ &= \boxed{\frac{P(B)P(A \mid B)}{P(A)}}. \end{aligned}$$

Conditioned Bayes' Rule: given events A, B, C,

$$P(A \mid B, C) = \frac{P(B \mid A, C)P(A \mid C)}{P(B \mid C)}.$$

Law of Total Probability

Let $B_1, \ldots, B_n$ be n disjoint events whose union is the entire sample space. Then, for any event A,

$$
\begin{aligned}
P(A) &= \sum_{i=1}^{n} P(A \cap B_i) \\
&= \sum_{i=1}^{n} P(A \mid B_i) P(B_i).
\end{aligned}
$$

We can then write Bayes' Rule as:

$$
\begin{aligned}
P(B_k \mid A) &= \frac{P(B_k) P(A \mid B_k)}{P(A)} \\
&= \boxed{\frac{P(B_k) P(A \mid B_k)}{\sum_{i=1}^{n} P(A \mid B_i) P(B_i)}}.
\end{aligned}
$$

Chain Rule

For any n events $A_1, \dots, A_n$, the joint probability can be expressed as a product of conditionals:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_2 \cap A_1) \cdots$$
$$P(A_n \mid A_{n-1} \cap A_{n-2} \cap \cdots \cap A_1).$$

Independence

Events A, B are independent if

$$P(AB) = P(A)P(B).$$

We denote this as $A \perp B$. From this, we know that if $A \perp B$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

Implication: If two events are independent, observing one event does not change the probability that the other event occurs.

In general: events $A_1, \ldots, A_n$ are **mutually independent** if

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Random Variable

A **random variable** X is a variable that probabilistically takes on different values. It maps outcomes to real values.

Probability Mass Function (PMF)

Given a discrete RV X, a PMF maps values of X to probabilities.

$$p_X(x) := p(x) := P(X = x)$$

For a valid PMF,

$$\sum_{x \in \mathrm{Val}(X)} p_X(x) = 1.$$

Cumulative Distribution Function (CDF)

A CDF maps a continuous RV to a probability (i.e. $\mathbb{R} \to [0, 1]$)

$$F_X(a) := F(a) := P(X \leq a).$$

A CDF must fulfill the following:

- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$
- If $a \leq b$, then $F_X(a) \leq F_X(b)$ (i.e. CDF must be nondecreasing)

Also note:

$$P(a \leq X \leq b) = F_X(b) - F_X(a).$$

Probability Density Function (PDF)

PDF of a continuous RV is simply the derivative of the CDF.

$$f_X(x) := f(x) := \frac{dF_X(x)}{dx}.$$

Expectation

Let g be an arbitrary real-valued function.

- If X is a discrete RV with PMF p_X:

$$\mathbb{E}[g(X)] := \sum_{x \in \mathrm{Val}(X)} g(x) p_X(x)$$

- If X is a continuous RV with PDF f_X:

$$\mathbb{E}[g(X)] := \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

Intuitively, expectation is a weighted average of the values of $g(x)$, weighted by the probability of x.

Markov's Inequality

Let X be a non-negative RV. Then for any $\tau > 0$,

$$P(X > \tau) \leq \frac{E[X]}{\tau}.$$

Properties of Expectation

For any constant $a \in \mathbb{R}$ and arbitrary real function f:

- $\mathbb{E}[a] = a$
- $\mathbb{E}[af(X)] = a\mathbb{E}[f(X)]$

Linearity of Expectation
Given n real-valued functions $f_1(X), \ldots, f_n(X)$,

$$\mathbb{E}\left[\sum_{i=1}^{n} f_i(X)\right] = \sum_{i=1}^{n} \mathbb{E}[f_i(X)].$$

Variance

The **variance** of a RV X measures how concentrated the distribution of X is around its mean.

$$\begin{aligned} \mathrm{Var}(X) &:= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2. \end{aligned}$$

Interpretation: $\mathrm{Var}(X)$ is the expected deviation of X from $\mathbb{E}[X]$.
Properties: For any constant $a \in \mathbb{R}$, real-valued function $f(X)$

- $\mathrm{Var}[a] = 0$
- $\mathrm{Var}[af(X)] = a^2 \mathrm{Var}[f(X)]$

Example Distributions

Distribution	PDF or PMF	Mean	Variance
Bernoulli(p)	$\begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, 1, \ldots, n$	np	$np(1-p)$
Geometric(p)	$p(1-p)^{k-1}$ for $k = 1, 2, \ldots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Uniform(a, b)	$\frac{1}{b-a}$ for all $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Gaussian(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for all $x \in (-\infty, \infty)$	μ	σ^2
Exponential(λ)	$\lambda e^{-\lambda x}$ for all $x \geq 0, \lambda \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Joint and Marginal Distributions

- Joint PMF for discrete RV's X, Y:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

  Note that

$$\sum_{x \in \mathrm{Val}(X)} \sum_{y \in \mathrm{Val}(Y)} p_{XY}(x, y) = 1.$$

- Marginal PMF of X, given joint PMF of X, Y:

$$p_X(x) = \sum_y p_{XY}(x, y).$$

Joint and Marginal Distributions

- Joint PDF for continuous RV's $X_1, \ldots, X_n$:

$$f(x_1, \ldots, x_n) = \frac{\partial^n F(x_1, \ldots, x_n)}{\partial x_1 \partial x_2 \cdots \partial x_n}$$

  Note that

$$\int_{x_1} \int_{x_2} \cdots \int_{x_n} f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n = 1.$$

- Marginal PDF of X_1, given joint PDF of $X_1, \ldots, X_n$:

$$f_{X_1}(x_1) = \int_{x_2} \cdots \int_{x_n} f(x_1, \ldots, x_n)\, dx_2 \cdots dx_n.$$

Expectation for multiple random variables

Given two RV's X, Y and a function $g : \mathbb{R}^2 \to \mathbb{R}$ of X, Y,

- for discrete X, Y:

$$\mathbb{E}[g(X, Y)] := \sum_{x \in \mathrm{Val}(X)} \sum_{y \in \mathrm{Val}(Y)} g(x, y) p_{XY}(x, y)$$

- for continuous X, Y:

$$\mathbb{E}[g(X, Y)] := \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y)\, dx\, dy.$$

Covariance

Intuitively: measures how much one RV's value tends to move with another RV's value. For RV's X, Y:

$$\begin{aligned}\mathrm{Cov}[X, Y] &:= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

- If $\mathrm{Cov}[X, Y] < 0$, then X and Y are negatively correlated
- If $\mathrm{Cov}[X, Y] > 0$, then X and Y are positively correlated
- If $\mathrm{Cov}[X, Y] = 0$, then X and Y are uncorrelated

Variance of two variables

$$\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}[X, Y]$$

Conditional distributions for RVs

Works the same way with RV's as with events:

- For discrete X, Y:

$$p_{Y|X}(y \mid x) = \frac{p_{XY}(x, y)}{p_X(x)}$$

- For continuous X, Y:

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

- In general, for continuous $X_1, \ldots, X_n$:

$$f_{X_1|X_2,\ldots,X_n}(x_1 \mid x_2, \ldots, x_n) = \frac{f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n)}{f_{X_2,\ldots,X_n}(x_2, \ldots, x_n)}.$$

Bayes' Rule for RVs

Also works the same way for RV's as with events:

- For discrete X, Y:

$$p_{Y|X}(y \mid x) = \frac{p_{X|Y}(x \mid y) p_Y(y)}{\sum_{y' \in \text{Val}(Y)} p_{X|Y}(x \mid y') p_Y(y')}$$

- For continuous X, Y:

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y) f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x \mid y') f_Y(y') \, dy'}.$$

Random Vectors

Given n RV's $X_1, \ldots, X_n$, we can define a random vector X s.t.

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}.$$

Note: all the notions of joint PDF/CDF will apply to X.
Given $g : \mathbb{R}^n \to \mathbb{R}^m$, we have:

$$g(X) = \begin{bmatrix} g_1(X) \\ g_2(X) \\ \vdots \\ g_m(X) \end{bmatrix}, \qquad \mathbb{E}[g(X)] = \begin{bmatrix} \mathbb{E}[g_1(X)] \\ \mathbb{E}[g_2(X)] \\ \vdots \\ \mathbb{E}[g_m(X)] \end{bmatrix}.$$

Covariance Matrices

For a random vector $X \in \mathbb{R}^n$, we define its covariance matrix Σ as the $n \times n$ matrix whose ij-th entry contains the covariance between X_i and X_j.

$$\Sigma = \begin{bmatrix} \mathrm{Cov}[X_1, X_1] & \cdots & \mathrm{Cov}[X_1, X_n] \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}[X_n, X_1] & \cdots & \mathrm{Cov}[X_n, X_n] \end{bmatrix}.$$

applying linearity of expectation and the fact that $\mathrm{Cov}[X_i, X_j] = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])]$, we obtain

$$\Sigma = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^T].$$

Properties:

- Σ is symmetric and PSD
- If $X_i \perp X_j$ for all i, j, then $\Sigma = \mathrm{diag}(\mathrm{Var}[X_1], \ldots, \mathrm{Var}[X_n])$

Multivariate Gaussian

The multivariate Gaussian $X \sim \mathcal{N}(\mu, \Sigma)$, $X \in \mathbb{R}^n$:

$$p(x; \mu, \Sigma) = \frac{1}{\det(\Sigma)^{1/2}(2\pi)^{n/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

Gaussian when $n = 1$.

$$p(x; \mu, \sigma^2) = \frac{1}{\sigma(2\pi)^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

Notice that if $\Sigma \in \mathbb{R}^{1\times 1}$, then $\Sigma = \mathrm{Var}[X_1] = \sigma^2$, and so $\Sigma^{-1} = 1/\sigma^2$ and $\det(\Sigma)^{1/2} = \sigma$.

Information and Entropy

Entropy

The entropy of a discrete random variable X is (we take 2 as the base of *log* operator)

$$H(X) = \mathbb{E}\Big[\mathcal{I}(X)\Big] = -\sum_{x\in\mathcal{X}} p(x)\log p(x).$$

Entropy measures the average uncertainty in X.

$$0 \leq H(X) \leq \log|\mathcal{X}|,$$

with maximum entropy when X is uniformly distributed.

Joint and Conditional Entropy

Joint entropy

The uncertainty associated with two random variables is

$$H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y).$$

Conditional entropy

The uncertainty remaining in Y after observing X is

$$H(Y \mid X) = -\sum_{x,y} p(x, y) \log p(y \mid x).$$

Mutual Information

Definition

Mutual information measures how much knowing one random variable reduces uncertainty about another:

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Equivalent expressions

$$I(X;Y) = H(X) + H(Y) - H(X,Y),$$

$$I(X;Y) = H(Y) - H(Y \mid X) = H(X) - H(X \mid Y).$$

Cross-entropy

$$H(p, q) = -\sum_x p(x) \log q(x)$$

for two distributions p and q.

Kullback–Leibler divergence

$$D_{\mathrm{KL}}(p \| q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

- $H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$.
- $D_{\mathrm{KL}}(p \| q) \geq 0$.
- In general, $D_{\mathrm{KL}}(p \| q) \neq D_{\mathrm{KL}}(q \| p)$.
- Cross-entropy is commonly used as a classification loss.

Examples

Suppose X is a fair binary random variable: $P(X=0) = P(X=1) = \frac{1}{2}$. Therefore, $H(X) = 1$ bit.

Case 1: $Y = X$

Knowing X completely determines Y:

$$H(Y \mid X) = 0,$$

$$I(X;Y) = H(Y) = 1 \text{ bit}.$$

Case 2: X and Y independent

Knowing X provides no information about Y:

$$H(Y \mid X) = H(Y) = 1,$$

$$I(X;Y) = 0.$$

Mutual information measures the reduction in uncertainty produced by observing another variable.

Thank You!

Questions?